

Parallel Database Architectures

1 Evolution of Architectures

The architecture of a parallel database system defines the relationship between processors, memory, and disks. Choosing the right architecture involves balancing **inter-processor communication efficiency** with **system scalability**.

2 Shared Memory Architecture

In a **shared-memory architecture**, multiple processors (or cores) share a common memory space and a common set of disks. Communication between processors is extremely efficient as they can exchange data simply by reading and writing to shared memory.

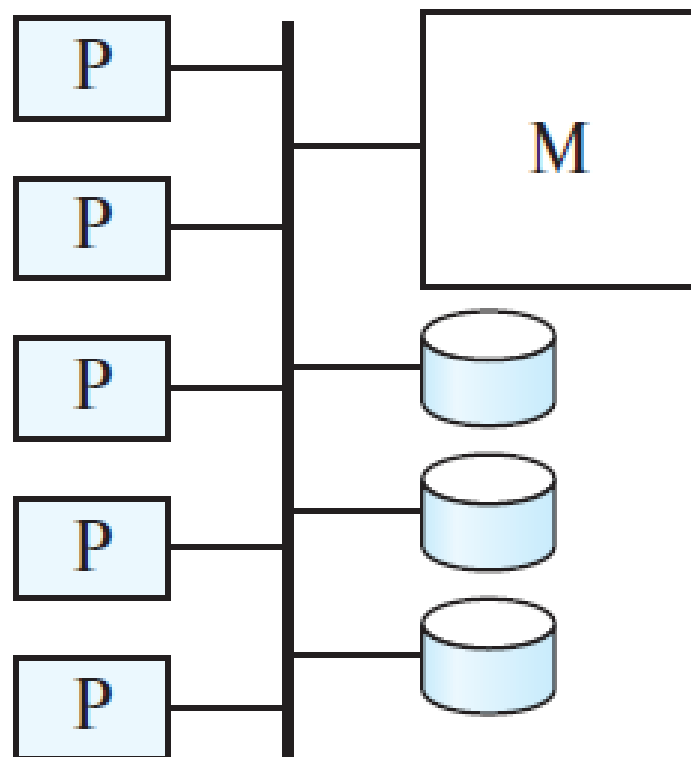


Figure 1: Shared Memory Architecture.

- **High Efficiency:** Data sharing is extremely fast, bypassing network protocols.
- **Scalability Limit:** Typically not scalable beyond 64 to 128 cores. The interconnection network between memory and processors becomes a major bottleneck.

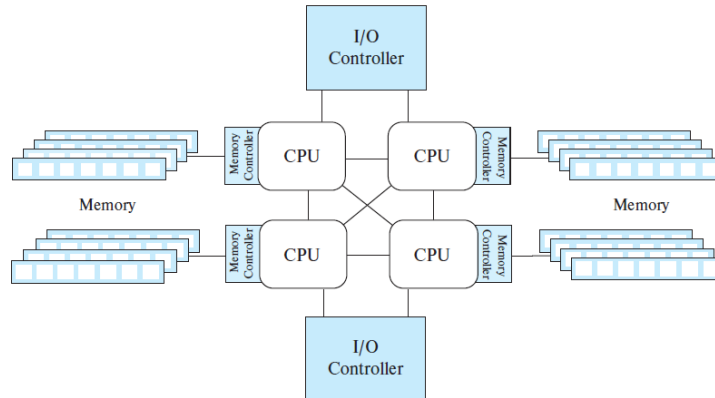


Figure 2: Modern Shared Memory

2.1 Cache Coherency & Memory Barriers

Modern multi-core processors use multiple cache levels. **Cache Coherency** ensures that all processors see up-to-date data. In **weak consistency** models, specialized hardware instructions are required:

Core 0	Core 1	Core 2	Core 3
L1 Cache	L1 Cache	L1 Cache	L1 Cache
L2 Cache	L2 Cache	L2 Cache	L2 Cache
Shared L3 Cache			

Figure 3: Cache Coherency and Memory Barriers.

- **sfence (Store Barrier):** Flushes cached data to memory.
- **lfence (Load Barrier):** Processes pending invalidations.
- **mfence:** A full memory fence performing both.

3 Shared Disk Architecture

All processors have their own **private memory** but share access to all disks via a high-speed interconnection network (often a **Storage Area Network - SAN**).

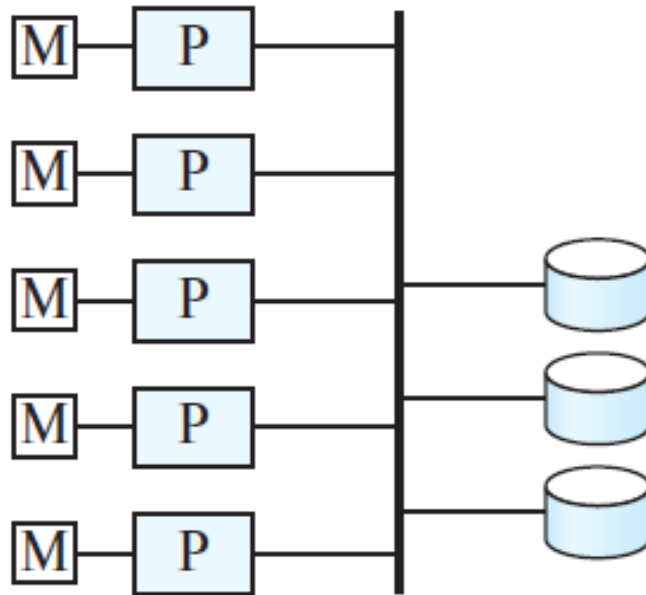


Figure 4: Shared Disk Architecture.

Fault Tolerance

This architecture is highly resilient. If a processor fails, its tasks can be assumed by another processor since the underlying data resides on shared disks accessible to all.

4 Shared Nothing Architecture

Each node is a self-contained unit consisting of its own processor, memory, and disks. There is no sharing of resources; all interaction happens via a high-speed interconnection network.

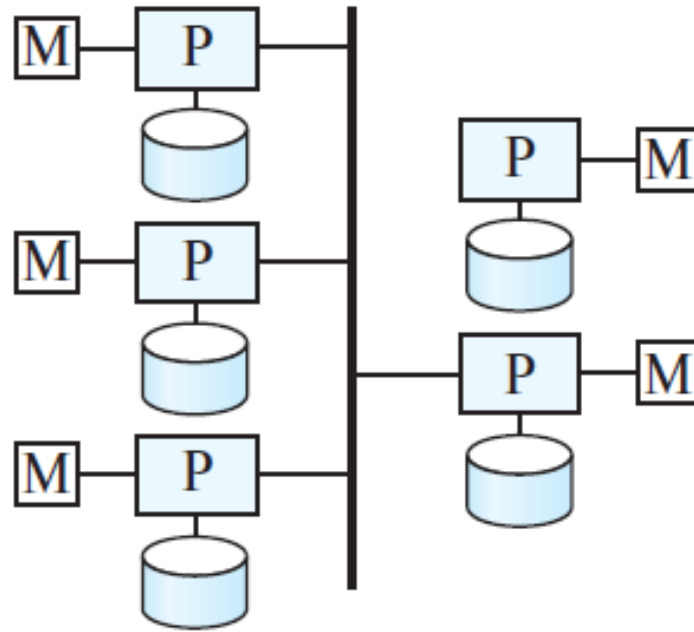


Figure 5: Shared Nothing Architecture.

- **Linear Scalability:** Can scale to thousands of nodes without resource interference.
- **Communication Cost:** Data movement involves higher overhead due to software interaction at both ends (sending/receiving node).

5 Hierarchical Architectures

Hierarchical (or hybrid) architectures combine the strengths of different models to manage massive scale while retaining local efficiency.

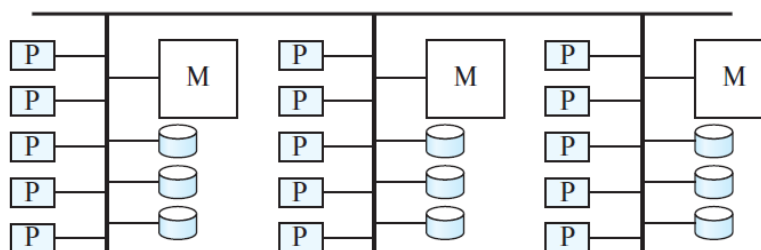


Figure 6: Hierarchical Architecture.

- **Typical Implementation:** A large-scale **Shared Nothing** cluster where each individual node is internally a **Shared Memory** multi-processor system.
- **Shared Disk Variant:** Alternatively, the top level could be a shared-disk system where each node is a multi-core shared-memory system.

6 Modern Abstractions

NUMA vs. RDMA

- **NUMA (Non-Uniform Memory Architecture):** A shared-memory software view of hardware that physically looks like shared-nothing. Locality of access is critical for performance.
- **RDMA (Remote Direct Memory Access):** Provides a shared-memory abstraction on shared-nothing systems by allowing one node to bypass another's CPU to access its RAM directly.

Design Warning

Careless programming with RDMA or NUMA can lead to severe performance degradation if data locality is ignored or if there are frequent remote memory references.